

A Design Framework For Equitable Wearables

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ABSTRACT

Wearables have become ubiquitous technologies that allow people to capture rich data to track their health information, immerse themselves in virtual reality, and stay connected. However, there is a consensus that wearables are only available to a select few due to cost, reducing the potential benefits they can provide to underserved populations. To ensure that wearables are designed more equitably and include the voices of underserved communities in the design process, I conducted a qualitative study with 19 participants from low-income communities and found that safety was a critical issue and that wearables can potentially be used to address their daily safety concerns. In this thesis position paper, I motivate the importance of codesigning wearables with and for underserved communities and propose a novel approach at the hardware and software levels for the development of safety-based wearable devices based on the needs of these communities.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Ubiquitous and mobile computing systems and tools*.

KEYWORDS

Inclusive wearables, Spatial Sensing, Low-power sensing, Safety

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1 PROBLEM STATEMENT

Over the past three decades, wearable research has enhanced human activity sensing and perception capabilities in the natural and virtual worlds opening new methods to gather rich data from on-body computations. However, this research community still needs to acknowledge how current wearable technologies can be useful or harmful for individuals in marginalized groups. Marginalized

groups face additional barriers and systemic structures of oppression, and excluding them from the design of wearables further marginalizes these groups of their use [5, 8].

As a first step toward designing more inclusive and equitable wearables, we conducted a qualitative study with 19 participants from minority low-income communities in two metropolitan cities in the United States to understand how members of these communities perceive wearables and what are the barriers and facilitators toward adoption. In my paper, Equityware [6], our findings show that safety is a critical issue among participants. Unfortunately, current wearable technologies on the market centered on safety do not meet the needs of participants because they are too pricey, flashy, have poor battery-life, too intrusive or do not take into consideration the dangerous geographies in which participants live in. Participants mentioned that most attacks occur in dark and dangerous places. The participants discussed that having a device that can **increase their spatial awareness** to detect if someone will attack them would be helpful.

In my thesis work, I propose to extend Equityware and build a wearable device that will utilize low-power hardware and software consisting of a highly efficient acoustic machine learning pipeline to augment a user's spatial perception. My research addresses the following:

- (1) Develop a framework to understand low-SES community members' barriers and facilitators towards adopting wearables. (Completed)
- (2) Understand how wearables can be designed to support the needs of low-SES communities. (Completed)
- (3) Explore novel spatial sensing techniques at the hardware and software levels while addressing the unique requirements of low-income communities. (In progress)

I will conduct a bench-marking campaign of the hardware, signal processing and machine learning pipeline, evaluating energy use and battery lifetime, accuracy and precision of the model, memory overhead, and performance, all across multiple (simulated) real-world scenarios in a free-living environment. I aim to develop a prototypical wearable device that will utilize acoustic sensing and a highly efficient sound recognition machine learning pipeline to increase a user's situational awareness. To ensure that the wearable is helpful to the community, I intend to deploy, evaluate, and refine this wearable with members of low-income communities through an iterative co-design user study. From the perspective of Ubicomp computer systems, my work bridges the gap between low-power, high-performance machine learning and acoustic sensing techniques with the existing constraints low-income communities face in terms of connectivity, power, and form factor revealed in the studies conducted in my previous work. Finally, this work contributes to the Ubicomp research community by introducing a social justice lens and the value of understanding the needs of

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communities that have traditionally been left out of technology design.

2 RELATED WORK

2.1 Co-Designing Wearables with Underserved Populations

Previous work has shown the value of co-designing technologies with and for low-SES communities [7, 13, 18]. Harrington et al. [12] examined the barriers African-American and Latinx communities in the U.S. face when encountering city technologies and generated solutions to these challenges but also warned of the unintentional harms technology can cause when lived experiences of marginalized groups are not taken into account.

Although these works have opened the door to more inclusive technological design, these works have only looked at co-designing with low-SES communities through information system approaches (mobile applications and dashboards). I extend the field by co-designing, building, and deploying physical wearables with low-SES communities through a hardware and signal processing approach.

2.2 Current Safety Wearables

Wearable tools for safety are currently available on the market. For example, wearables that are targeted for safety consist of panic buttons to send distress signals (She's Birdie, Invisawear), share locations with people you trust (Arlo Safe App and Button), or trigger fake phone calls (Flare bracelet). Though the idea of carrying a gadget that does the above for you, in a real scenario, sometimes these features are not enough because it relies on the user's reactivity, which may sometimes need to be faster. Furthermore, these devices are impractical as they can be expensive, may require monthly data plans, and are not sustainable at times. For instance, the battery life of Invisawear only lasts one year and costs \$99 to replace.

The Ubicomp and Wearables research community are at the forefront of creating novel sensing technologies to augment human perception, yet it has failed to create technologies for critical situations such as safety. We, as a community, can do better. To help move the research community forward, my work aims to find creative ways that put user needs at the center and build a wearable that is affordable, sustainable, unobtrusive, and able to increase users' awareness of their spatial surroundings.

2.3 Acoustic and Spatial Sensing

Augmenting human perception through acoustic sensing has been implemented in areas such as interactivity for touch interfaces [15, 17], enabling spatial interaction through smart watches [11, 16], and sound activity recognition [14, 22]. Most recent work in spatial acoustic sensing have enabled new possibilities to collect spatial information from ubiquitous acoustic devices [4, 9]. Although these innovations show great promise in using spatial sensing of acoustic signals in other fields such as VR, speech recognition, drone localization, and many others, the utility of low-power techniques for accurate machine learning on high-data rate acoustic signals has not been explored in the context of safety. My proposed work will focus on examining whether spatial sensing of acoustic signals could be useful in the context of safety and if it can be prove to be an alternative medium to develop wearables that are minimally

invasive, inexpensive, and sustainable while still being able to enhance a users spatial awareness to react much faster to a dangerous situation.

3 WORK DONE SO FAR

Over the past two years, I have worked on research aimed at: 1) understanding how members of low-SES communities perceive wearable devices and 2) understand how wearables can be designed to support the needs of low-SES communities [6]. This work has helped motivate my proposed work.

3.1 Methodology

Between December 2021 and March 2022, I performed an exploratory study to understand the needs of low-SES communities that influence the adoption and use of wearable devices. I conducted a series of semi-structured interviews with 19 participants who are members of low-income communities from two metropolitan cities (Los Angeles and Chicago).

The study was broken down into two phases. An initial round of semi-structured interviews revealed a strong focus on safety concerns. This informed a second round of data collection that focused on participants' perspectives, reactions, and recommendations for the design of safety-based wearables. In the second phase of the study I engaged in co-design and an elicitation activity using storyboards [19, 23]. The storyboards allowed me to introduce the concept of safety wearables, elicit what participants think safety wearables could or should do, suggest feedback on current features found in safety-related wearable devices, and explore prospective future applications and or comfort with the technology.

Storyboards were motivated by initial interviews and highly commercialized existing wearables around safety. We selected a number of features based on preliminary discussions and existing safety-based wearables that would provide a landscape of wearables which participants could challenge and investigate, drawing on their lived experience. We note that we did find that these storyboards got participants talking about ideas not mentioned in the stories themselves.

3.2 Evaluation

Audio recordings of the interviews were transcribed, resulting in a total of 21 hours of interviews. Data from both studies were combined and analyzed together. My research team and I analyzed the data using a grounded theory approach [3]. I and two co-authors performed open coding on the transcripts and identified themes; we reviewed the transcripts and collaboratively discussed associated codes to look for consistencies and differences in the data.

This work revealed a strong interest in using wearables for safety among members of low-SES communities. We learned that participants experience daily safety concerns such as gang violence, lack of infrastructure and services, and gender-based violence. Participants expressed interest in having wearable devices that address these safety concerns. Table 1 shows the features that participants felt were helpful and harmful found in current wearables, as well as the features proposed for future wearables. A notable feature that the participants proposed was spatial sensing, aimed at increasing the wearer's sense of spatial surroundings. Participants used their

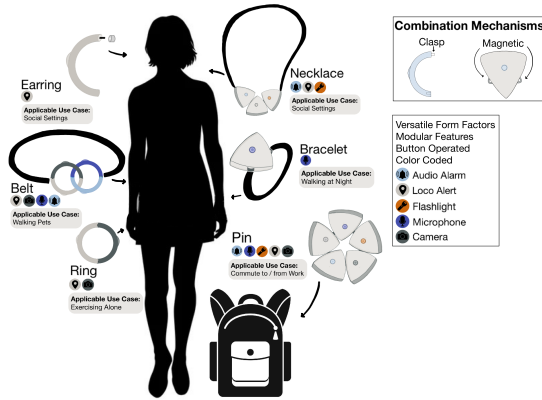


Figure 1: This figure shows the outcomes from the storyboards, and our attempt to distill the design requirements, context, and conversations over the course of the study into a cohesive wearable system for multi-faceted safety in low-SES communities. By providing a mechanism that affords wearing in different forms and on varied body parts (i.e., necklace versus ring versus backpack), a user can have the ability to re-configure their device at will, to adjust and navigate tensions around stigma, visibility, as well as adapt to requirements for various situations.

lived experience to imagine features they want to see in future wearables and is the inspiration behind my proposed work.

3.3 Contribution: Equityware

To address the critical barriers and harmful features found in existing wearables, I set a research agenda, which I term "Equityware", focused on co-designing inclusive technologies with and for marginalized communities and highlighting opportunities for HCI, UbiComp, and Wearables researchers to collaborate and contribute at three levels: hardware, software, and research & education. We designed a prototype system (shown in Figure 1) that could integrate the major themes and features participants suggested and overcome many barriers to usage participants presented. The core of the design embraces the idea of **contextual reconfigurability** in hardware and software. This study is the first of its kind to work on co-designing safety wearables directly with individuals living in low-SES communities and shows that there is great value in working with these individuals to understand user needs and broaden the potential affordances wearables can offer. This work contributes to my proposed work by providing guidelines for developing inclusive future wearables.

4 WORK PROPOSED

Moving forward, for my dissertation, I propose developing a safety-based wearable device co-designed with and for members of low-SES communities based on the principles of my Equityware research

agenda, which serves as a prototypical demonstration. I will explore novel spatial sensing techniques at the hardware and software levels while addressing the unique requirements of low-income communities. For the device, I aim to develop novel low-power hardware capable of augmenting a user's spatial surroundings to alert them if a person or object is within their vicinity using acoustic sensing. I propose to develop a sound recognition machine learning pipeline to enable human activity recognition algorithms combined with sensors, such as microphones, that require minimal interface interaction. To be clear, this work is not meant to be a contribution that is purely software or hardware; the contribution is developing a co-designed platform aimed at leveraging existing technologies for a specific and novel application with a population that "has been traditionally left out of the technological design process" in the past.

4.1 Methodology

My approach for developing this wearable device will be through a series of iterative co-design processes where I build hardware and software prototypes and receive feedback from my user group, which will consist of members from low-SES communities. This is important because the community that this wearable aims to serve has unique constraints that affect the hardware, software, and user studies. I explain the potential avenues that I intend to explore to address these constraints below.

4.1.1 Hardware. The hardware design of my proposed wearable must function with constrained power not only for sustainability purposes but because of the physical restrictions that community members face, such as not being able to power a device due to frequent home power outages. Furthermore, this wearable device would be useless if it had poor battery life, not to mention that it would increase the cost of replacing the battery every year.

For this reason, I will explore different types of low-power microphones that can capture relevant data in different areas of the body and different real-world spaces (loud vs. quiet spaces). For the first prototype I am using the VM2020 MEMS microphone [1] an updated version used by Garg et al. [9]. Through iterative co-design user studies, I will learn if there are other physical constraints I must be aware of which will influence and narrow down the choice of microphone. My choice of microphone will then influence the type of data I can collect and, therefore, inform the machine-learning pipeline with which I can experiment. Knowing which ML models are useful would be crucial because then I can know if my hardware platform will require MCUs with CNN accelerators such as the Max78000 [2] or if the physical constraints are too much to handle, I may need to choose a different MCU.

4.1.2 Software. The size of the machine learning model I use will be limited by the physical factors of the hardware I can use. Running user studies will allow me to understand how large I can make the model, how much training data I need, and how much data I can reuse from existing datasets. I would have to optimize all these parameters around the Equityware space. I intend to use the model used by Garg in [9] as an initial model and explore other existing acoustic sensing models [10, 20, 21] and edge device ML models (tinyML, tensor flow lite, MobileNetV2) on which I can build. Lastly,

Safety Concerns	Helpful Features	Harmful features & Critical Barriers	Proposed Features
Gang violence Lack of infrastructure Gender-based violence	Camera (automatic evidence capture) Spatial sensing Small size Versatility Ultra-low power or batteryless	Privacy invasive cameras Racial Discrimination Lack of design control Affordability Flashiness Short battery-life	Location Sharing Microphone/Voice recognition Flashlight Night vision camera

Table 1: This table presents participants’ safety concerns, the critical barriers/harmful features of wearable devices we presented to them in the storyboards, the accepted and valuable features, and participants’ proposed features.

for potential outputs for the model we use, we envision a quadrant based model that would predict the presence or no presence for quadrants around a person. The predicted quadrant would guide actuation via vibro motors, phone alerts, or another mechanism.

4.1.3 Iterative Co-design. In the summer of 2023, I expect to run a series of iterative co-design studies to determine the physical constraints that will influence my choices of hardware components I can use. I plan to recruit participants through social media channels such as Facebook and Instagram, as I did in my previous work. Eligible participants would be invited to meet in a preselected safe space to help them feel comfortable and prevent any power imbalances between them and me as the lead researcher.

After consenting, participants will engage in a Wizard of Oz study designed to mimic scenarios where physical barriers and constraints could occur. Participants will complete online surveys after each session, in which they will be asked questions akin to "Do you feel this is a scenario you would experience in reality? What design choices would you make to this existing prototype to make it more comfortable? Are there other scenarios or constraints we missed that should have been addressed? I expect to run two studies; the first will test our initial prototype. Based on the feedback from the first sessions, I will iteratively design and apply the necessary changes to my prototype. In the second round, the wearable device will be deployed in the wild, where participants will take the device home and use it for a week. After the in-wild study, participants will return the device and complete an exit survey where they will report their feedback to me and discuss their likes and dislikes. Once all data is gathered and ready to report, I aim to publish my results.

4.2 Evaluation

I will evaluate the wearable that I build across the hardware, software, and user study. For the hardware I will benchmark the device’s energy efficiency, size, weight, cost, comfort, and usability. I will measure its battery life to see how long it lasts and how many times it needs to be recharged.

On the software side, I will evaluate what triggers an event. I will examine my models accuracy, precisions, and truth table and compare that to a ground truth device. Because I will be exploring a design space, I will pick out interesting features from the machine learning algorithms used by performing an ablation study and see which features are relevant to what I aim to predict with my model. In situations where an individual feels they are being followed at

night in an urban landscape, where there are many echoes, the highest rated feature can alert the user quickly that someone is behind them from a certain distance and is measured through, for example, a kurtosis of the frequency bands of axle. Of course, this is just an example of a potential evaluation process. Finally, for the user study, participant reactions, comfortability, limitations, and recommendations will be gathered through survey responses.

4.3 Contribution

The co-design of a safety-based wearable device aimed at empowering individuals from low-income communities to address their safety concerns using novel spatial sensing techniques will be the first of its kind. Further, I layout Equityware as a research agenda to encourage and facilitate greater work in the space of building wearable devices and hardware with and for traditionally overlooked communities. Equityware is the first step in creating a long term research agenda and I recognize the need for this research to be scalable across different parts of the world. Rather than adopting a "one size fits all" approach, I advocate for a meta-approach where I first study a specific population to learn valuable lessons as a research community. This will enable us to understand how wearable technology requirements may differ for low-SES individuals in various parts of the world. Future research question to consider in this work are 1) what are the lower boundaries that can define the bare minimum of these safety wearables with a certain degree of reliability and data quality? Often, wearables need a companion device to offload data and computation capabilities. Can we think of smart wearables that don’t rely on such companion devices, can handle certain tasks on-device, and can be used as personal safety accessories rather than an accessory to another device, which compounds the entire cost? I aim to explore this in future iterations of my work.

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